SMART INDIA HACKATHON 2026

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UnifyMat — One Nation, One Material Code

- **Problem Statement ID – SIH26099**
- **Problem Statement Title- AI-Driven Standardization and Harmonization of Material Codes Across CPSEs**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team Name (Registered on portal)- Hawkeye**

Hawkeye

UnifyMat
One Nation, One Material Code

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❖ Proposed Solution

- One National Material Code (NMC) per real material, all CPSEs
- Same bolt today: 5 codes, 5 prices (₹16–₹21) — merged safely
- AI proposes · hard facts veto · humans approve

Proven on 403 records · 5 CPSEs: precision 1.000 · 0 wrong merges · ~10 s on CPU

5 CPSE material masters → one national code

CPCL | IOCL | NTPC | SAIL | GAIL → ONE NATIONAL MATERIAL CODE NMC-FAST-BOLT-9BF0ABAC

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TECHNICAL APPROACH

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- Tech: Python, Streamlit, MiniLM embeddings, TF-IDF fallback, RapidFuzz
- Flow: normalize → 35 attrs → match & veto → deterministic NMC + audit

1 · NORMALIZE
~35 attributes from free text

→

2 · MATCH & VETO
embeddings rank · hard facts decide

→

3 · ISSUE NMC
legacy mapping + audit trail

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FEASIBILITY AND VIABILITY

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- Feasible: any ERP CSV in; ~10 s per run
- Risk: wrong merges — veto + officer gate
- Proved on ground-truthed synthetic data

403 → 264
national codes issued

0
trap violations (8.8 vs 10.9)

7,951
unsafe pairs auto-rejected

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IMPACT AND BENEFITS

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- 300+ CPSEs on one national material language
- Safety: 8.8 vs 10.9 mix-up now impossible

Economic: 12.8% avg price spread exposed — pump ₹26,674 vs ₹54,000 (51%); aggregated demand → national rate contracts

1.000 auto-merge precision	**0** wrong merges	**89.7%** recall with approvals

Every merge is officer-approved — named, timestamped, fully auditable.

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RESEARCH AND REFERENCES

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- Problem Statement SIH26099 — MoP&NG / CPCL

- Reimers & Gurevych (2019) — Sentence-BERT sentence embeddings · arXiv:1908.10084
- all-MiniLM-L6-v2 model · huggingface.co/sentence-transformers/all-MiniLM-L6-v2
- Papadakis et al. (2020) — Blocking & Filtering for Entity Resolution · ACM CSUR
- ISO 898-1 — fastener property classes 8.8 / 10.9 (the grade trap in our data)
- Government e-Marketplace (GeM) · gem.gov.in — national public-procurement context
- Working prototype, benchmark + verified outputs · github.com/Varunsai1930/SIH